GENERAL PROJECTIONS FROM POINTS OFF X RESTRICT TO FINITE MORPHISMS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original reference 01WC: sources/stacks-chow/tags.txt, source lines 2477–2477. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01W6: sources/stacks-chow/tags.txt, source lines 2471–2471. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01WN: sources/stacks-chow/tags.txt, source lines 2488–2488. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02OG: sources/stacks-chow/tags.txt, source lines 3515–3515. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let V be a vector space of dimension $n+1$. Let $X \subset \mathbf{P}(V)$ be a closed subscheme. If $X \neq \mathbf{P}(V)$, then there is a nonempty Zariski open $U \subset \mathbf{P}(V)$ such that for all closed points $p \in U$ the restriction of the projection r_p defines a finite morphism $r_p|_X : X \to \mathbf{P}(W_p)$.*

Proof. We claim the lemma holds with $U = \mathbf{P}(V) \setminus X$. For a closed point p of U we indeed obtain a morphism $r_p|_X : X \to \mathbf{P}(W_p)$. This morphism is proper because X is a proper scheme (Morphisms, Lemmas 01WC and 01W6). On the other hand, the fibres of r_p are affine lines as can be seen by a direct calculation. Hence the fibres of $r_p|X$ are proper and affine, whence finite (Morphisms, Lemma 01WN). Finally, a proper morphism with finite fibres is finite (Cohomology of Schemes, Lemma 02OG). □

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